

Extended Infusion Anti-Pseudomonal Beta-Lactams

SOP 119-006

VA Ann Arbor Healthcare System
Ann Arbor, MI 48105

Signatory Authority:

Ginny L. Creasman, Pharm.D., FACHE
Medical Center Director

Service Line(s):

Pharmacy

Responsible Owner:

Ron Kendall, PharmD
Clinical Pharmacy Specialist, Infectious
Disease

Effective Date:

May 1, 2024

Recertification Date:

May 1, 2029

1. Purpose and Authority

This medical center SOP establishes authority for pharmacists to independently change the infusion time of certain IV beta-lactams. Pharmacokinetic/pharmacodynamic principles should be used to optimize antimicrobial utilization whenever possible. Beta-lactam antibiotics have time-dependent bacterial killing and therefore their efficacy is increased when the concentration at the site of infection is above the minimum inhibitory concentration of the bacteria for a longer duration. Prolonging the infusion time is a recommended strategy which increases the likelihood of successful antimicrobial activity. In certain cases, lower total daily doses of the antibiotic can be used if given via extended infusion which can mitigate drug toxicities and cost.

2. Procedures

a. Pharmacist The pharmacist is responsible for:

(1) Changing the infusion time of the intravenous antibiotic at order verification as indicated in the chart below:

	First Dose	Subsequent Dose Infusion Time based on the Dosing Interval	
	-	q6h	q8H, q12H, q24H
Cefepime, Ceftazidime	30 minutes	180 minutes	240 minutes
Meropenem, Imipenem/Cilastatin			
Piperacillin/Tazobactam			

(a) The first dose of an antibiotic should be the standard 30-minute infusion in order to not delay time to an initial therapeutic concentration in a patient with a severe infection.

(b) Patients with poor renal function or on renal replacement therapy may still benefit from extended infusion and are not excluded.

(c) IV antibiotics being dispensed for home infusion are excluded as it is not practical to extended infusion at home without an automated pump.

(2) Changing the start time of the subsequent scheduled doses to begin half-way into the first dosing interval (i.e., for cefepime 2g IV q8h, give 2g IV over 30 minutes now and start cefepime 2g IV q8h via 4-hour infusion 4 hours after the initial dose).

b. Prescriber. The prescriber is responsible for:

(1) Ordering the appropriate antibiotic and dose considering the specific parameters of the patient (height, weight, renal function, liver function, albumin, etc.) and the infection (bacteria, minimum inhibitory concentration, penetration of the antibiotic to the site of infection, severity of infection, therapeutic index and potential toxicities of the antimicrobial, etc).

(2) Contact infectious disease provider for any questions regarding appropriate antibiotic selection or dose.

3. OTHER PARAGRAPHS

Nothing in this document precludes the use of different infusion times being selected on a case-by-case basis by the provider in conjunction with the infectious diseases team, nor the use of extended or continuous infusion of any other IV antibiotic not specifically listed in this policy.

4. DEFINITIONS

None.

5. REFERENCES

None.

6. REVIEW

Minimal review annually, revision history maintained via service master SOP Spreadsheet

7. RECERTIFICATION

This SOP is scheduled for recertification on or before the last working day of [Month Year – 5 years from effective date]. In the event of contradiction with national policy, the national policy supersedes and controls.

8. SIGNATORY AUTHORITY

GINNY L. CREASMAN, Pharm.D., FACHE
Medical Center Director

Date Approved: Month Day, Year

NOTE: *The signature remains valid until rescinded by an appropriate administrative action.*

Policy Memorandum Distribution Tracking Sheet

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REMARKS All records regardless of format (e.g., paper, electronic, electronic systems) created by this policy must be managed as required by the National Archives and Records Administration approved records schedules found in VHA Records Control Schedule 10-1.